

# Smit Chaudhary

QUANTUM ALGORITHMS DEVELOPER · QUANTUM PHYSICIST · SOFTWARE ENGINEER

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## Experience

### Quantum Algorithms Developer

Amsterdam, The Netherlands

PASQAL

Nov 2022 - Present

- Developed Quantum Scientific Machine Learning (QSciML) algorithms. Built and trained variational quantum circuits to solve industrially relevant aerodynamics PDEs, creating quantum alternatives to fully classical physics-informed neural networks (PINNs)
- Served as technical lead on European Union funded projects, translating quantum research into practical solutions and driving business development opportunities with major aerospace industry partners
- Architected quantum software infrastructure and built hardware-aware circuit capabilities, making strategic technical decisions that aligned algorithms with neutral atom hardware constraints while enhancing both open-source and proprietary frameworks
- Mentored 3 research interns and 2 junior developers, leading to successful project contributions and 3 research publications

### Quantum Computing Intern

California, USA

MENTEN AI

May 2022 - Sept 2022

- Developed a novel fully quantum generative adversarial network (QGAN) architecture for discrete and binary data generation
- Investigated and implemented quantum circuit features such as noise reuploading in the generator and auxiliary qubits in the discriminator to enhance expressivity
- Demonstrated the model's performance on synthetic datasets and low-energy states of Ising models, showing successful data reproduction and generalization capabilities

## Education

### Technische Universiteit Delft

Delft, The Netherlands

MASTER OF SCIENCE IN APPLIED PHYSICS

August 2020 - July 2022

- **Thesis:** Quantum Walk Based Algorithms for Qubit Mapping (**Supervisor:** Prof. Sebastian Feld)
- **Research Experience:** Honors Project on Noise Induced Barren Plateaus at Leiden University (**Supervisor:** Prof. Jordi Tura)
- **Relevant Coursework:** Applied Quantum Algorithms, Quantum Information, Quantum Computing Architecture

### Indian Institute of Technology, Kanpur

Kanpur, India

BACHELOR OF SCIENCE IN PHYSICS

July 2016 - May 2020

- **Undergraduate Project:** Bohmian Mechanics and Quantum Information (**Supervisor:** Prof. Kaushik Bhattacharya)
- **Research Experience:** Summer Research at IISER Kolkata on Quantum Machine Learning (**Supervisor:** Prof. Prasanta Panigrahi)
- **Relevant Coursework:** Quantum Computing, Quantum Field Theory, Statistical Mechanics, Probability and Statistics, Optics

## Publications

### Weak Form Regularisation for Solving Differential Equations with Quantum Neural Networks

A. Paine, **S. Chaudhary**, A. A. Gentile

*Submitted to Quantum Techniques in Machine Learning, 2025*

Addresses key limitations in Quantum PINNs where traditional point based approaches fail at boundary propagation. Combines local collocation points and global weak form terms to improve accuracy and stability.

### Solving Fluid Dynamics Equations with Differentiable Quantum Circuits

**S. Chaudhary**, G. T. Balducci, O. Kyriienko, P. K. Barkoutsos, L. Cardarelli, A. A. Gentile

*Proceedings of the 35th Parallel CFD International Conference 2024, May 2025*

Demonstrates applications of hybrid quantum-classical algorithms to solve complex fluid dynamics problems. Shows how data integration and quantum feature maps offer advantage over classical methods.

## Quantum Circuit Training with Growth-Based Architectures

C. Duffy, **S. Chaudhary**, G. V. Velikova

*arXiv preprint*, Nov 2024

Develops novel growth-based training strategies for parametrized quantum circuits that dynamically adjust circuit depth during optimization to manage model complexity to prevent overfitting.

## Towards a scalable discrete quantum generative adversarial neural network

**S. Chaudhary**, P. Huembeli, I. MacCormack, T. L. Patti, J. Kossaifi, A. Galda

*Quantum Science and Technology*, April 2023

Introduces a novel fully quantum (and connected) generative adversarial network with both quantum generator and discriminator, incorporating noise reuploading and auxiliary qubits.

## Quantum Machine Learning: A Review and Current Status

N. Mishra, M. Kapil, H. Rakesh, et al., **S. Chaudhary**, et al.

*Data Management, Analytics and Innovation, Springer Singapore*, Sept 2020

Reviews the current status (as of 2020) and previous literature on Quantum Machine Learning (QML)

## Research Projects

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### Quantum Walk-Based Algorithms for Qubit Mapping

TU Delft, Delft, The Netherlands

**ADVISOR:** PROF. SEBASTIAN FELD

August 2021 - May 2022

- Developed a quantum walk based algorithm for qubit mapping in quantum circuits, framed as a combination of a **Constraint Satisfaction Problem (CSP)** and an optimization problem
- Implemented a modified quantum algorithm for speeding up **backtracking search** in CSPs to reduce qubit requirements by pre-computing mapping order and using efficient ancilla register management
- Designed a classical greedy routing algorithm to minimize overhead in terms of **SWAP** operations for sliced quantum circuits, optimizing the path of qubit mappings across circuit segments
- Evaluated the algorithm's performance against increasing circuit depth and number of qubits, showing an **overhead ratio ('R') of approximately 1.1** for deep and large circuits, indicating efficient routing
- Analyzed the algorithmic complexity, demonstrating a space complexity of  **$O(n \log n)$**  for the quantum walk algorithm, specifically optimizing for qubit count in the implementation

### Quantum Generative Adversarial Networks

Leiden University, Leiden, The Netherlands

APPLIED QUANTUM ALGORITHMS COURSE

February 2021 - April 2022

- Implemented a fully quantum GAN with both quantum generator (**QCBM**) and quantum discriminator for quantum state generation
- Developed a custom quantum state vector simulation framework using **NumPy** matrix operations, creating modular classes for gates, circuits and GAN components
- Implemented quantum gradient computation by deriving circuit construction rules for parametric gate derivatives, enabling back-propagation in quantum circuits, enabling efficient training through custom gradient descent implementation
- Validated predictions showing Wasserstein loss convergence to zero and fidelity approaching unity during adversarial training
- Demonstrated successful quantum state reproduction achieving **fidelity >0.99** on 2-qubit systems, converging to target states

### Barren Plateaus in QNN Training with Correlated Noise

Leiden University, Leiden, The Netherlands

**ADVISOR:** PROF. JORDI TURA

April 2021 - April 2022

- Validated established Barren Plateau theory through implementation of Haar random circuits with local Pauli noise
- Implemented realistic quantum hardware noise models for small shallow circuits including correlated noise channels
- Analyzed the impact of noise correlation and strength on Barren Plateau formation beyond uncorrelated noise regimes
- Pursued improved theoretical bounds for Barren Plateau onset under correlated noise scenarios

## Skills

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### Programming

Python, C/C++, Rust, Verilog

### Languages

English, Hindi, Gujarati

### Libraries & Utilities

Git, PyTorch, NumPy, PennyLane, Tensorflow, Jax, Cirq, Qiskit